

Quasi-optimum distance flag codes

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WCC 2026

- 1 Constant dimension codes
- 2 Flag codes
- 3 Quasi-ODFCs
- 4 A construction

Outline

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- 2 Flag codes
- 3 Quasi-ODFCs
- 4 A construction

Notation

Let...

- q be a **prime power**,
- $\mathbb{F}_q$ be the **finite field** with q elements,
- $n \geq 2$
- $\mathbb{F}_q^n$ is the n -dimensional **vector space** over $\mathbb{F}_q$,
- $\mathcal{G}_q(k, n)$, the **Grassmannian** of dimension $1 \leq k \leq n - 1$.

Constant dimension codes

Given two subspaces $\mathcal{U}, \mathcal{V} \in \mathcal{G}_q(k, n)$, their **subspace distance**

$$d_S(\mathcal{U}, \mathcal{V}) = 2(k - \dim(\mathcal{U} \cap \mathcal{V})).$$

Definition

A **constant dimension code** $\mathcal{C}$ in $\mathcal{G}_q(k, n)$ is a nonempty collection of k -dimensional subspaces of $\mathbb{F}_q^n$. Its **minimum distance** is

$$d_S(\mathcal{C}) = \min\{d_S(\mathcal{U}, \mathcal{V}) \mid \mathcal{U}, \mathcal{V} \in \mathcal{C}, \mathcal{U} \neq \mathcal{V}\}.$$

Constant dimension codes with maximum distance

If $\mathcal{C} \subseteq \mathcal{G}_q(k, n)$, then $d_S(\mathcal{C})$ is an even integer

$$0 \leq d_S(\mathcal{C}) \leq \begin{cases} 2k & \text{if } 2k \leq n, \\ 2(n-k) & \text{if } 2k \geq n. \end{cases}$$

Spread codes

Definition

A **k -spread code** $\mathcal{S}$ of $\mathbb{F}_q^n$ is a constant dimension code in $\mathcal{G}_q(k, n)$ such that every nonzero vector of $\mathbb{F}_q^n$ lies on one, and only one, subspace in $\mathcal{S}$.

Properties:

- k -spread codes exist $\Leftrightarrow k$ divides n ,
- $k \leq \frac{n}{2}$,
- $d_{\mathcal{S}}(\mathcal{S}) = 2k$ is **maximum**,
- $|\mathcal{S}| = \frac{q^n - 1}{q^k - 1}$.

Definition

For $k \leq \frac{n}{2}$, a code $\mathcal{C} \subseteq \mathcal{G}_q(k, n)$ is called **partial k -spread** if $d_S(\mathcal{S}) = 2k$.

Definition

A set $\mathcal{C} \subseteq \mathcal{G}_q(k, n)$ such that

$$\mathcal{U} \cap \mathcal{V} = \mathcal{X} \text{ for every } \mathcal{U}, \mathcal{V} \in \mathcal{C}, \mathcal{U} \neq \mathcal{V}$$

is called a **sunflower** of center $\mathcal{X}$.

$$d_S(\mathcal{C}) = 2(k - c) \text{ with } c = \dim(\mathcal{X}).$$

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Flags

Definition

Given integers $1 \leq t_1 < \dots < t_r \leq n-1$, a **flag** of type $(t_1, \dots, t_r)$ on $\mathbb{F}_q^n$ is a sequence of $\mathbb{F}_q$ -subspaces

$$\mathcal{F} = (\mathcal{F}_1, \dots, \mathcal{F}_r)$$

such that

- $\mathcal{F}_1 \subsetneq \mathcal{F}_2 \subsetneq \dots \subsetneq \mathcal{F}_r$.
- $\dim(\mathcal{F}_i) = t_i$, for every $1 \leq i \leq r$

$\mathcal{F}_q((t_1, \dots, t_r), n) \rightarrow$ the **flag variety** of type $(t_1, \dots, t_r)$ on $\mathbb{F}_q^n$.

The flag distance

Definition

Given $\mathcal{F} = (\mathcal{F}_1, \dots, \mathcal{F}_r)$ and $\mathcal{F}' = (\mathcal{F}'_1, \dots, \mathcal{F}'_r)$ flags of type $(t_1, \dots, t_r)$, their **flag distance** is

$$d_f(\mathcal{F}, \mathcal{F}') = \sum_{i=1}^r d_S(\mathcal{F}_i, \mathcal{F}'_i).$$

Flag codes

Definition

A **flag code** of type $(t_1, \dots, t_r)$ on $\mathbb{F}_q^n$ is a nonempty subset

$$\mathcal{C} \subseteq \mathcal{F}_q((t_1, \dots, t_r), n).$$

Its **minimum distance** is

$$d_f(\mathcal{C}) = \min\{d_f(\mathcal{F}, \mathcal{F}') \mid \mathcal{F}, \mathcal{F}' \in \mathcal{C}, \mathcal{F} \neq \mathcal{F}'\}$$

 D. Liebhold, G. Nebe and A. Vazquez-Castro,
[Network Coding with Flags](#),
 Designs, Codes and Cryptography,
 Vol. 86(2) (2018), 269–284.

Flag codes

Bounds for the flag distance

Given a flag code $\mathcal{C} \subseteq \mathcal{F}_q((t_1, \dots, t_r), n)$, then

$$0 \leq d_f(\mathcal{C}) \leq D = 2 \left(\sum_{2t_i \leq n} t_i + \sum_{2t_i > n} (n - t_i) \right).$$

Definition

A flag code $\mathcal{C} \subseteq \mathcal{F}_q((t_1, \dots, t_r), n)$ with

$$d_f(\mathcal{C}) = D.$$

is called **optimum distance flag code** (ODFC).

Projected codes

Definition

Let $\mathcal{C} \subseteq \mathcal{F}_q((t_1, \dots, t_r), n)$ be a flag code. For every $1 \leq i \leq r$, the set

$$\mathcal{C}_i = \{\mathcal{F}_i \in \mathcal{G}_q(t_i, n) \mid (\mathcal{F}_1, \dots, \mathcal{F}_i, \dots, \mathcal{F}_r) \in \mathcal{C}\}$$

is called the i -projected code of $\mathcal{C}$.

Definition

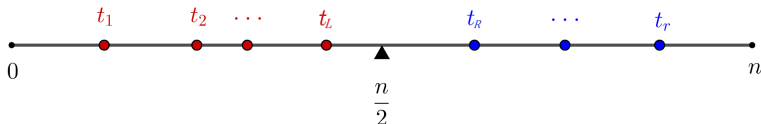
A flag code $\mathcal{C} \subseteq \mathcal{F}_q((t_1, \dots, t_r), n)$ is said to be disjoint if

$$|\mathcal{C}| = |\mathcal{C}_1| = \dots = |\mathcal{C}_r|.$$

Characterization of ODFCs

Theorem (Alonso-González, N.-P. and Soler-Escrivà, 2020)

The code $\mathcal{C} \subseteq \mathcal{F}_q((t_1, \dots, t_r), n)$ is an **ODFC** if, and only if, $\mathcal{C}$ is **disjoint** and $d_S(\mathcal{C}_i)$ is **maximum** for **every** $1 \leq i \leq r$.



Theorem (N.-P. and Soler-Escrivà, 2022)

The code $\mathcal{C} \subseteq \mathcal{F}_q((t_1, \dots, t_r), n)$ is an **ODFC** if, and only if, $|\mathcal{C}| = |\mathcal{C}_L| = |\mathcal{C}_R|$ and $d_S(\mathcal{C}_i)$ is **maximum** for $i \in \{L, R\}$.

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Quasi-optimum distance flag codes

Definition

A flag code $\mathcal{C} \subseteq \mathcal{F}_q((t_1, \dots, t_r), n)$ is called **quasi-optimum distance flag code** (QODFC) if

$$d_f(\mathcal{C}) = D - 2.$$

They can be **disjoint** or **not**.

Projected codes of higher length

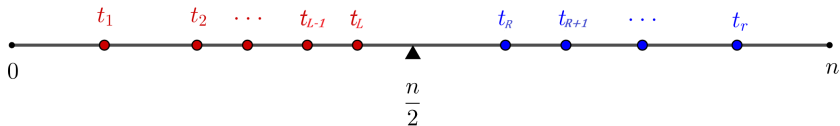
Definition

Let $\mathcal{C} \subseteq \mathcal{F}_q((t_1, \dots, t_r), n)$ be a flag code. For every $1 \leq i_1 < \dots < i_M \leq r$, the set

$$\mathcal{C}_{(i_1, \dots, i_M)} = \{(\mathcal{F}_{i_1}, \dots, \mathcal{F}_{i_M}) \mid \mathcal{F} \in \mathcal{C}\}$$

is called the **projected (flag) code** of $\mathcal{C}$ of type $(t_{i_1}, \dots, t_{i_M})$.

The disjoint case



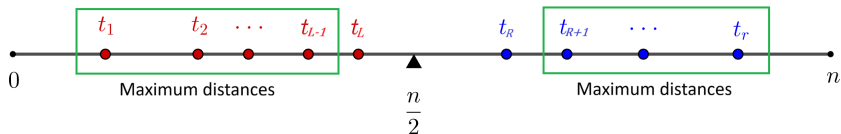
Theorem (Alonso-González, N.-P. 2026)

Let $\mathcal{C} \subseteq \mathcal{F}_q((t_1, \dots, t_r), n)$ be a **disjoint** flag code.

$\mathcal{C}$ is a **QODFC** if, and only if,

- 1 $\mathcal{C}_{(L,R)}$ is a **QODFC** and
- 2 $d_S(\mathcal{C}_i)$ is **maximum** for every $i \notin \{L, R\}$.

The disjoint case



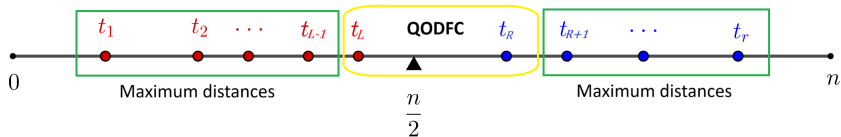
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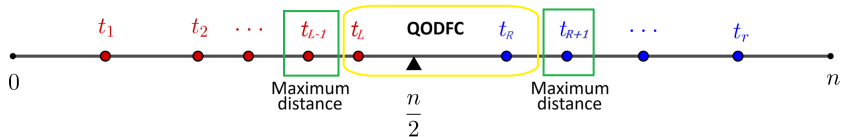
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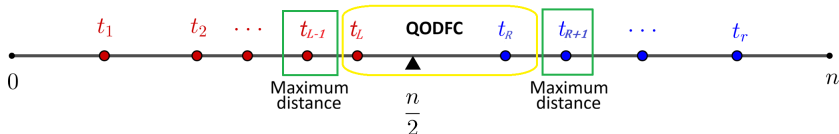
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The disjoint case



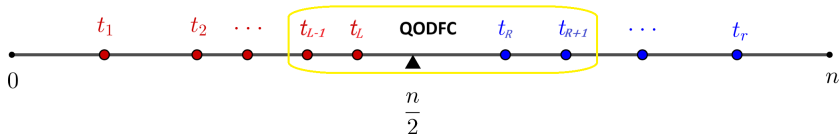
$(t_{L-1}, t_L, t_R, t_{R+1})$ the **distinguished** (sub)type vector

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Let $\mathcal{C} \subseteq \mathcal{F}_q((t_1, \dots, t_r), n)$ be a **disjoint** flag code. Then

$\mathcal{C}$ is a **QODFC** $\iff \mathcal{C}_{(L-1, L, R, R+1)}$ is a **QODFC**.

The disjoint case



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The non-disjoint case

Non-disjoint:

$\exists \mathcal{F}, \mathcal{F}' \in \mathcal{F} \mid \mathcal{F} \neq \mathcal{F}' \text{ but } \mathcal{F}_i = \mathcal{F}'_i \text{ for some } 1 \leq i \leq r.$

Equivalently: $|\mathcal{C}_i| < |\mathcal{C}|.$

Proposition

Let $\mathcal{C} \subseteq \mathcal{F}_q((t_1, \dots, t_r), n)$ be a **non-disjoint QODFC**, then $|\mathcal{C}_i| < |\mathcal{C}|$ happens **only** for dimensions $t_i \in \{1, n-1\}.$

The non-disjoint case

Non-disjoint:

$$\exists \mathcal{F}, \mathcal{F}' \in \mathcal{F} \mid \mathcal{F} \neq \mathcal{F}' \text{ but } \mathcal{F}_i = \mathcal{F}'_i \text{ for some } 1 \leq i \leq r.$$

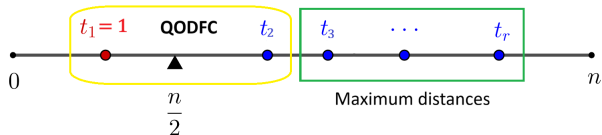
$$\text{Equivalently: } |\mathcal{C}_i| < |\mathcal{C}|.$$

Theorem (Alonso-González, N.-P. 2026)

Let $\mathcal{C} \subseteq \mathcal{F}_q((t_1, \dots, t_r), n)$ be a **non-disjoint QODFC**, then one of the following statements holds:

- ① $t_1 = 1$ and $t_2 > \frac{n}{2}$. Moreover: $|\mathcal{C}_1| < |\mathcal{C}_2| = \dots = |\mathcal{C}_r| = |\mathcal{C}|$.
- ② $t_r = n - 1$ and $t_{r-1} < \frac{n}{2}$ with $|\mathcal{C}| = |\mathcal{C}_1| = \dots = |\mathcal{C}_{r-1}| > |\mathcal{C}_r|$.
- ③ Type $(1, n - 1)$ and $|\mathcal{C}_i| < |\mathcal{C}|$, for $i \in \{1, 2\}$.

The non-disjoint case

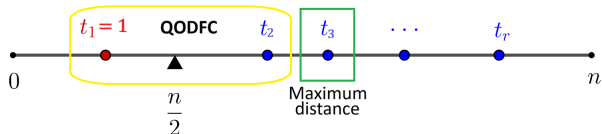


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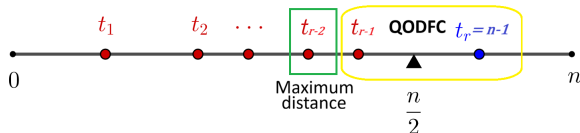


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The non-disjoint case

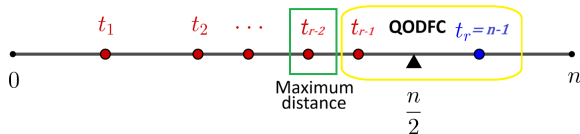


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The ingredients

- For the **disjoint** case,
- $n = 2k + 1$ (**but similar for $n = 2k + 2$**),
- a **hyperplane** $\mathcal{X} \in \mathcal{G}_q(2k, 2k + 1)$,
- a **k -spread** $\mathcal{S} = \{\mathcal{S}_1, \dots, \mathcal{S}_{q^{k+1}}\}$ of $\mathcal{X}$,
- **generator matrices** $A_i \in \mathbb{F}_q^{k \times n}$ such that $\mathcal{S}_i = \text{rowsp}(A_i)$,
- $A_i^{(j)}$ first j rows of A_i ,
- $\mathcal{S}_i^{(j)} = \text{rowsp}(A_i^{(j)})$,
- $u \in \mathbb{F}_q^n \setminus \mathcal{X}$.

The projected codes

For every $1 \leq j \leq k$:

We consider the **partial spread**

$$\mathcal{S}^{(j)} = \{\mathcal{S}_1^{(j)}, \dots, \mathcal{S}_{q^k+1}^{(j)}\}$$

of dimension j and size $q^k + 1$. (**Maximum distance**)

For every $1 \leq j \leq k$:

$$\mathcal{S}^{(j)} \oplus \langle u \rangle = \{\mathcal{S}_i^{(j)} \oplus \langle u \rangle \mid 1 \leq i \leq q^k + 1\}.$$

is a **sunflower** of dimension $j + 1$ and center $\langle u \rangle$ and size $q^k + 1$.
(**Maximum distance minus two**).

The projected codes

For higher dimensions...

The subspaces

$$\left\{ \begin{array}{lcl} \mathcal{U}_1^{(j)} & = & \mathcal{S}_1 \oplus \mathcal{S}_2^{(j-1)} \oplus \langle u \rangle, \\ \mathcal{U}_2^{(j)} & = & \mathcal{S}_2 \oplus \mathcal{S}_3^{(j-1)} \oplus \langle u \rangle, \\ \vdots & \vdots & \vdots \\ \mathcal{U}_{q^k+1}^{(j)} & = & \mathcal{S}_{q^k+1} \oplus \mathcal{S}_1^{(j-1)} \oplus \langle u \rangle, \end{array} \right.$$

have dimension $k + j$, for every $1 \leq j \leq k$.

The projected codes

We form:

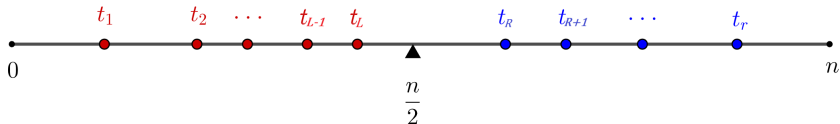
$$\mathbf{C}^{(j)} = \{\mathcal{U}_1^{(j)}, \dots, \mathcal{U}_{q^k+1}^{(j)}\} \subseteq \mathcal{G}_q(k+j, 2k+1)$$

for $1 \leq j \leq k$.

Theorem (Alonso-González, N.-P. 2026)

For every $1 \leq j \leq k$, the code $\mathbf{C}^{(j)} \subseteq \mathcal{G}_q(k+j, 2k+1)$ has **maximum distance**.

For $n = 2k + 1$ and type $(t_1, \dots, t_r)$:

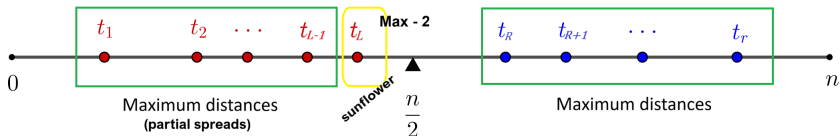


$$\mathcal{F}^i = (\mathcal{F}_1^i, \dots, \mathcal{F}_L^i, \dots, \mathcal{F}_r^i)$$

with subspaces

$$\mathcal{F}_j^i = \begin{cases} \mathcal{S}_i^{(t_j)} & \text{if } 1 \leq j \leq L-1, \\ \mathcal{S}_i^{(t_{j-1})} \oplus \langle u \rangle & \text{if } j = L, \\ \mathcal{U}_i^{(t_{j-k})} = \mathcal{S}_i \oplus \mathcal{S}_{i+1}^{(t_{j-k-1})} \oplus \langle u \rangle & \text{if } L < j \leq r, \end{cases}$$

For $n = 2k + 1$ and type $(t_1, \dots, t_r)$:

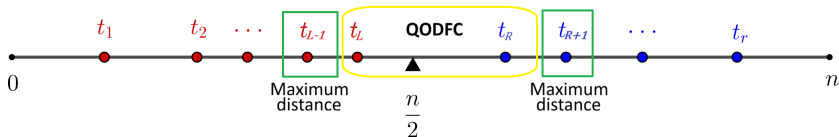


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The construction

Theorem (Alonso-González and N.-P., 2026)

The flag code

$$\mathcal{C} = \{\mathcal{F}^1, \dots, \mathcal{F}^{q^k+1}\}$$

is a **disjoint QODFC** of type $\mathbf{t} = (t_1, \dots, t_r)$ on $\mathbb{F}_q^n$ with size $q^k + 1$.

☰ C. Alonso-González and M.A. N.-P.,
Quasi-Optimum Distance Flag Codes,
Finite Fields and Their Applications,
Vol. 112 (2026), 102799.



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